WORK SHEET - 02 **VECTOR & 3D**

[HINTS & SOLUTIONS]

Q.1 $\sqrt{14}$

Sol. $\because$ Normal makes equal angles with coordinate axes. Therefore drs of normal can be taken as 1, 1, 1.
$\therefore$ Equation of the plane is $x + y + z = d$

Any point B on the line $\frac{x-1}{1} = \frac{y+2}{1} = \frac{z-3}{2}$ can be taken as $(\lambda + 1, \lambda - 2, 2\lambda + 3)$.

$\therefore$ drs of AB will be $(\lambda, \lambda - 4, 2\lambda)$
$\because$ AB is parallel to plane
$\therefore$ it is perpendicular to normal to plane.
$\therefore$ $B = (2, -1, 5)$
$\therefore$ $AB = \sqrt{1^2 + 3^2 + 2^2} = \sqrt{14}$.

Q.2 $\frac{-10}{3}$

Sol. Since $\vec{a}_1, \vec{a}_2$ & $\vec{a}_3$ are non-coplanar vectors
$\therefore$ $x + y - 3 = 0$ ......(i)
$2x - y + 2 = 0$ ......(ii)
$2x + y + \lambda = 0$ ......(iii)

From (i) & (ii) $x = \frac{1}{3}$, $y = \frac{8}{3}$

$\therefore$ from (ii) $\lambda = -\frac{10}{3}$

Q.3 $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$

Sol. Let direction ratios of the line be $<a, b, c>$, then
$2a - b + c = 0$
$a - b - 2c = 0$

i.e. $\frac{a}{3} = \frac{b}{5} = \frac{c}{-1}$

$\therefore$ direction ratios of the line are $<3, 5, -1>$
Any point on the line l is $(2 + \lambda, 2 - \lambda, 3 - 2\lambda)$. It lies on the plane Π if
$2(2 + \lambda) - (2 - \lambda) + (3 - 2\lambda) = 4$
i.e. $4 + 2\lambda - 2 + \lambda + 3 - 2\lambda = 4$
i.e. $\lambda = -1$
$\therefore$ The point of intesection of the line and the plane is $(1, 3, 5)$

$\therefore$ Equation of the required line is $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$

Q.4 $4x + 3y + 5z = 50$

Sol. Let a point $(3\lambda + 1, \lambda + 2, 2\lambda + 3)$ of the first line also lies on the second line.

Then $\dfrac{3\lambda+1-3}{1} = \dfrac{\lambda+2-1}{2} = \dfrac{2\lambda+3-2}{3} \quad \Rightarrow \quad \lambda = 1$

Hence the point of intersection P of the two lines is (4, 3, 5)

Equation of plane perpendicular to OP where O is (0, 0, 0) and passing through P is

$4x + 3y + 5z = 50$

Q.5 18

Sol. $\vec{V}_1 \cdot \vec{V}_2 = 0 \Rightarrow a + b + c = 0$

$$\begin{array}{ccccc} \dfrac{a}{-2} & \dfrac{b}{1} & \dfrac{c}{1} & \Rightarrow & \dfrac{3!}{2!} = 3 \\ -2 & 0 & 2 & \Rightarrow & 3! = 6 \\ -1 & 0 & 1 & \Rightarrow & 3! = 6 \\ -1 & -1 & 2 & \Rightarrow & \dfrac{3!}{2!} = 3 \end{array}$$

Total number of vectors = 18

Q.6 75

Sol. Centroid of tetrahedron having vertices (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) and (x_4, y_4, z_4) is

$$G \equiv \left(\frac{x_1+x_2+x_3+x_4}{4}, \frac{y_1+y_2+y_3+y_4}{4}, \frac{z_1+z_2+z_3+z_4}{4}\right)$$

So, $\dfrac{a+1+2+0}{4} = 1 \Rightarrow a = 1$

$\dfrac{2+b+1+0}{4} = 2 \Rightarrow b = 5$

$\dfrac{3+2+c+0}{4} = 3 \Rightarrow c = 7$

So, $a^2 + b^2 + c^2 = 1 + 25 + 49 = 75.$

Q.7 $(1, 1, \sqrt{2})$

Sol. Let the required plane has equation $ax + by + cz = d$

$\because$ Point (1, 0, 0) and (0, 1, 0) lies on it so $a = d$, $b = d$

So, equation of plane can be written as $ax + ay + cz = a$ ......(1)

$\because$ Plane (1) and plane (2) $x + y = 3$ make an angle of $\dfrac{\pi}{4}$,

So $\cos\dfrac{\pi}{4} = \left|\dfrac{a+a+0}{\sqrt{a^2+a^2+c^2}\,\sqrt{1^2+1^2}}\right|$

$\sqrt{2a^2+c^2} = 2a \Rightarrow 2a^2 + c^2 = 4a^2 \Rightarrow c = \pm\sqrt{2}\,a$

So, equation of plane can be written $ax + ay \pm \sqrt{2}\,az = a$

or $x + y \pm \sqrt{2}\,z = 1$

So, direction ratio are $(1, 1, \sqrt{2})$.

Q.8 $\sin^{-1}\sqrt{\sqrt{\frac{3}{2}}-1}$

Sol. $|\hat{v}_1\times(\hat{v}_2\times\hat{v}_3)|^2 = |(\hat{v}_1\cdot\hat{v}_3)\hat{v}_2-(\hat{v}_1\cdot\hat{v}_2)\hat{v}_3|^2$

$$\sin^2(90° - \theta)\cdot\sin^2\frac{\pi}{4} = \left|\left(\cos\frac{\pi}{6}\right)\hat{v}_2-\left(\cos\frac{\pi}{3}\right)\hat{v}_3\right|^2$$

$$\cos^2\theta\cdot\frac{1}{2} = \frac{3}{4}+\frac{1}{4}-2\cdot\frac{\sqrt{3}}{2}\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{2}}$$

$$\cos^2\theta = 2-\sqrt{\frac{3}{2}}$$

$$\theta = \cos^{-1}\left(\sqrt{2-\sqrt{\frac{3}{2}}}\right)$$

Q.9 $\frac{3}{2}$

Sol. $|\vec{a}\times\vec{b}| = \sqrt{|\vec{a}|^2|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2} = \sqrt{18-9} = 3$

$\therefore\ \left|(\vec{a}\times\vec{b})\times\vec{c}\right| = \left|\vec{a}\times\vec{b}\right|\,|\vec{c}|\sin 30^{o} = \frac{(3)\,(1)}{2} = \frac{3}{2}$ (As, $|\vec{c}-\vec{a}|^2 = 8 \Rightarrow |\vec{c}| = 1$)

Q.10 $\frac{1}{\sqrt{2}}$

Sol. $P_1 : 2x+y+z=1$ ... (i)
$P_2 : 3x+y+2z=2$ ... (ii)
(i) and (ii) represents planes. Sovle them to obtain equation of line.

$\Rightarrow\quad L_1 : \frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-1}{-1}$ ... (iii)

Also, $x=y=z \Rightarrow L_2 : \frac{x-0}{1} = \frac{y-0}{1} = \frac{z-0}{1}$ ... (iv)

$\therefore$ Shortest distance between them

$$= \left|\frac{\hat{k}\cdot\left((\hat{i}-\hat{j}-\hat{k})\times(\hat{i}+\hat{j}+\hat{k})\right)}{\left|(\hat{i}-\hat{j}-\hat{k})\times(\hat{i}+\hat{j}+\hat{k})\right|}\right| = \left|\frac{\hat{k}\cdot 2(\hat{k}-\hat{j})}{2\left|\hat{k}-\hat{j}\right|}\right| = \frac{1}{\sqrt{2}}$$

Q.11 0

Q.12 $\frac{1}{3}(3\vec{a}+4\vec{b}+8\vec{c})$

Sol. We are given that $\hat{x}+\hat{y}+\hat{z} = \vec{a}$ .....(I)

$\Rightarrow\ |\hat{x}+\hat{y}+\hat{z}|^2 = |\vec{a}|^2 \ \Rightarrow \sum \hat{x}.\hat{y} = \frac{1}{2}$

Let $\hat{x}.\hat{y} = p,\ \hat{y}.\hat{z} = q,\ \hat{z}.\hat{x} = r$

$\therefore\ p + q + r = \frac{1}{2}$ .....(1)

Also, $(\hat{x}.\hat{z})\hat{y} - (\hat{x}.\hat{y})\hat{z} = \vec{b} \Rightarrow r\hat{y} - p\hat{z} = \vec{b}$

and $(\hat{x}.\hat{z})\hat{y} - (\hat{y}.\hat{z})\hat{x} = \vec{c} \Rightarrow r\hat{y} - q\hat{x} = \vec{c}$ ... (3)

Take dot product with $\hat{x}$ in (I), we get

$$p + r = \frac{1}{2} \qquad \text{... (4)}$$

Take dot product with $\hat{y}$ in (I), we get

$$p + q = \frac{3}{4} \qquad \text{... (5)}$$

$\therefore\ p = \frac{3}{4},\ q = 0,\ r = \frac{-1}{4}$

Also, $\hat{x} = \frac{1}{3}(3\vec{a} + 4\vec{b} + 8\vec{c})$

$\hat{y} = -4\vec{c}$

$\hat{z} = \frac{4}{3}(\vec{c} - \vec{b})$

Q.13 110

Q.14 12

Sol. $\vec{v}_1 = (\sin\theta)\hat{i} + (\cos\theta)\hat{j} + (a-3)\hat{k}$

$\vec{v}_2 = (\sin\theta + \cos\theta)\hat{i} + (\cos\theta - \sin\theta)\hat{j} + (b-4)\hat{k}$

$\vec{v}_3 = (\cos\theta)\hat{i} - (\sin\theta)\hat{j} + (c-5)\hat{k}$

(i) Given, $\vec{v}_1 + \vec{v}_2 + \vec{v}_3 = \lambda\hat{i}$

$\Rightarrow\ (2\sin\theta + 2\cos\theta)\hat{i} + (2\cos\theta - 2\sin\theta)\hat{j} + (a+b+c-12)\hat{k} = \lambda\hat{i}$

$\therefore\ 2\cos\theta - 2\sin\theta = 0$ and $a + b + c - 12 = 0$

$\Rightarrow\ \tan\theta = 1$ and $a + b + c = 12;\ a, b, c \in N$

$\theta \in [0, 2\pi]$ $\quad\therefore\ a + b + c = 9$

$\therefore\ \theta = \frac{\pi}{4}$ or $\frac{5\pi}{4}$ $\qquad {}^{11}C_2 = 55$

$\therefore$ Number of quadruplets (a, b, c, θ) are $55 \times 2 = 110$.

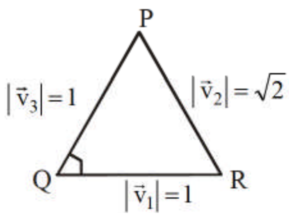
(ii) Given, $\vec{v}_1, \vec{v}_2, \vec{v}_3$ represents sides of ΔPQR in xy-plane

$\therefore$ Coefficeint of $\hat{k}$ in all 3 vector = 0

Hence, $a = 3$, $b = 4$ and $c = 5$

Also, Triangle PQR is right triangle

$$\Delta PQR = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$



and $\Delta ABC = \frac{1}{2} \times 3 \times 4 = \frac{12}{2}$

$$\therefore \quad \frac{\Delta ABC}{\Delta PQR} = \frac{\frac{12}{2}}{\frac{1}{2}} = 12$$

Q.15 $a + b = -1$

Sol. Plane perpendicular to σ_1 and σ_2 has direction ratios of normal

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix} = 16\left(-\hat{i} + 3\hat{j} + 2\hat{k}\right)$$

Also, point of intersections of lines L_1 and L_2 is $(5, -2, -1)$

$\therefore$ Required equation of plane is

$$-1\ (x - 5) + 3\ (y + 2) + 2(z + 1) = 0$$

$\Rightarrow \quad x - 3y - 2z = 13$

Q.16 $2x - 3y + z + 2\sqrt{14} = 0$

$2x - 3y + z - 2\sqrt{14} = 0$

Sol. Equation of plane through points A, B and C is $\begin{vmatrix} x-1 & y-1 & z-1 \\ 1 & 2 & 4 \\ -2 & -1 & 1 \end{vmatrix} = 0$

$\therefore$ Equation of the plane ABC is $2x - 3y + z = 0$

Let the equation of the required plane be $2x - 3y + z = k$, then $\left|\frac{k}{\sqrt{4+9+1}}\right| = 2$

$k = \pm 2\sqrt{14}$

$\therefore$ Equation of the required plane is $2x - 3y + z \pm 2\sqrt{14} = 0$

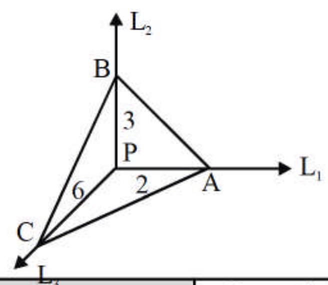
Q.17 $V = 6$ cubic units

$d = \frac{6}{\sqrt{14}}$ units

Sol. L_1, L_2 and L_3 are mutually perpendicular lines and all these lines are passing through P(1, 2, 3).

$$V = \frac{1}{6}\left[\overrightarrow{PA}\ \overrightarrow{PB}\ \overrightarrow{PC}\right] = \frac{1}{6} \times 2 \times 3 \times 6 = 6 \text{ cubic units.}$$

$$V = \frac{1}{3}(\text{Base area})(\text{Height})$$



$$6 = \frac{1}{3} \times \sqrt{\left(\frac{1}{2}\cdot 2\cdot 3\right)^2 + \left(\frac{1}{2}\cdot 3\cdot 6\right)^2 + \left(\frac{1}{2}\cdot 2\cdot 6\right)^2} \times d$$

$\therefore\ d = \dfrac{18}{\sqrt{9+81+36}} = \dfrac{18}{\sqrt{126}} = \dfrac{6}{\sqrt{14}}$ units

Aliter: Shifting origin at $(1, 2, 3)$ and considering lines L_1, L_2, L_3 as x-axis, y-axis, z-axis
Equation of plane A, B, C

$$\frac{x}{2}+\frac{y}{3}+\frac{z}{6}=1$$

$$3x + 2y + z - 6 = 0$$

Now, perpendicular distance of the plane from the origin

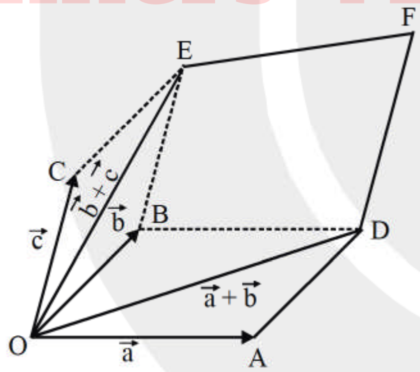
$$d = \frac{6}{\sqrt{9+4+1}} = \frac{6}{\sqrt{14}}$$

Q.18 in the plane of $\vec{a}$ and $\vec{b}$.
equally inclined to $\vec{a}$ and $\vec{b}$
perpendicular to $\vec{a} \times \vec{b}$

Sol. Obviously, $\vec{v} = \dfrac{\vec{a}}{|\vec{a}|} + \dfrac{\vec{b}}{|\vec{b}|}$ is a vector in the plane of $\vec{a}$ and $\vec{b}$ (by theorem in plane) and hence perpendicular to $\vec{a} \times \vec{b}$. Also, $\vec{v}$ is equally inclined to $\vec{a}$ and $\vec{b}$ as it is along the angle bisector.

Q.19 [5]

Sol. $\left|\vec{a}\times\vec{b}\right| = 6,\ \left|\vec{b}\times\vec{c}\right| = 10,\ \left|\vec{c}\times\vec{a}\right| = 8$



Area of quadrilateral BDFC = Area of paralleogram ODFE – (Area of ΔODB + Area of ΔOBE)

$S = \left|(\vec{a}+\vec{b})\times(\vec{b}+\vec{c})\right| - (3 + 5)$

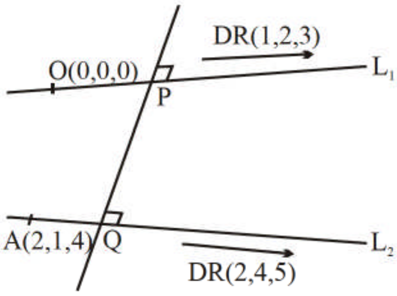
$= \left|\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}\right| - 8$

$= 24 - 8 = 16$

$\therefore\ \left[\dfrac{S}{3}\right] = 5$

Q.20 [6]

Sol. PQ distance is shortest distance between L_1 and L_2.



$$\text{S.D.} = \frac{\left|\left[2\hat{i}+\hat{j}+4\hat{k} \quad \hat{i}+2\hat{j}+3\hat{k} \quad 2\hat{i}+4\hat{j}+5\hat{k}\right]\right|}{\left|\left(\hat{i}+2\hat{j}+3\hat{k}\right)\times\left(2\hat{i}+4\hat{j}+5\hat{k}\right)\right|}$$

$$= \frac{3}{\left|-2\hat{i}+\hat{j}\right|} = \frac{3}{\sqrt{5}}$$

or $\quad (PQ)^2 = \dfrac{9}{5} \Rightarrow 5D = 9$

DRs of line L are proportional to $(-2, 1, 0)$.